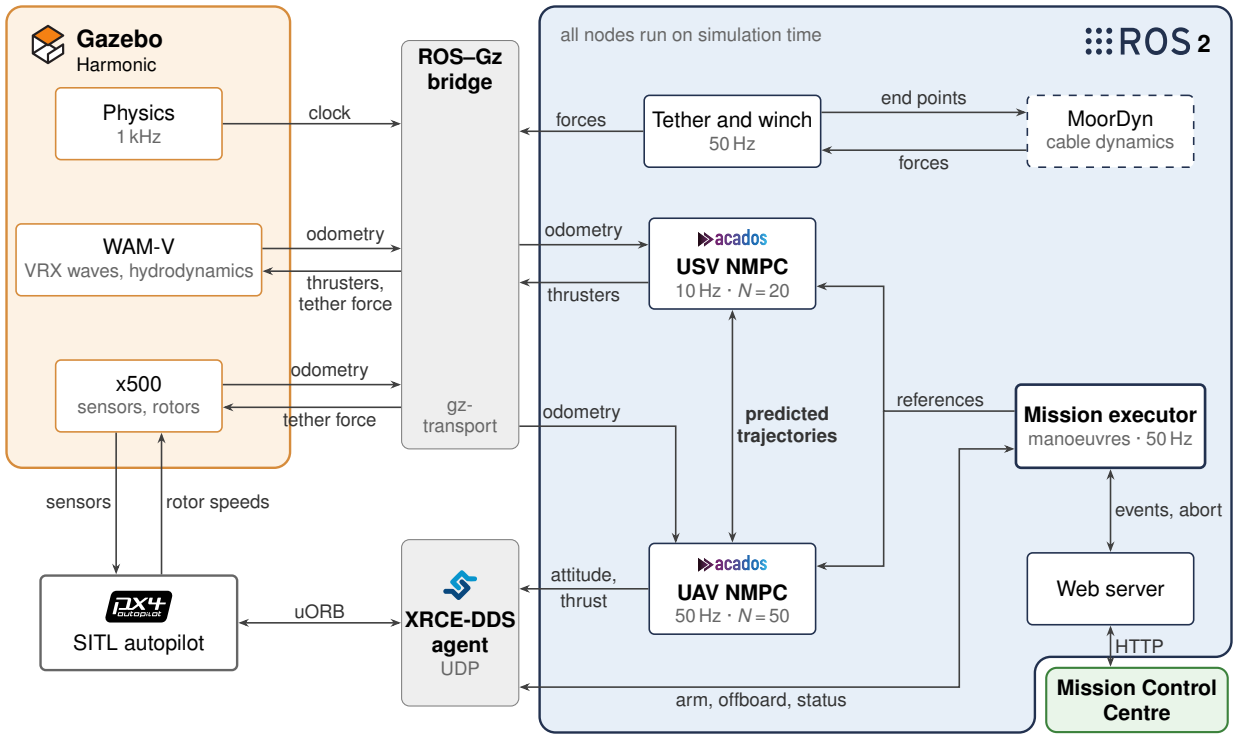

Gazebo
Harmonic
Physics
1 kHz
clock
WAM-V
VRX waves, hydrodynamics
odometry
thrusters, tether force
x500
sensors, rotors
odometry
tether force
sensors
rotor speeds
PX4 AUTOPILOT
SITL autopilot
uORB
ROS–Gz bridge
gz-transport
XRCE-DDS agent
UDP
all nodes run on simulation time
ROS 2
forces
Tether and winch
50 Hz
end points
forces
MoorDyn
cable dynamics
odometry
thrusters
acados
USV NMPC
10 Hz · N = 20
odometry
predicted trajectories
references
Mission executor
manoeuvres · 50 Hz
events, abort
attitude, thrust
acados
UAV NMPC
50 Hz · N = 50
Web server
HTTP
arm, offboard, status
Mission Control Centre